TASK 8: MACHINE LEARNING MODEL

Loan Prediction Dataset

Objective:
Build a prediction model

Tasks:

- Data preprocessing
- Feature selection
- Train/test split
- Model training

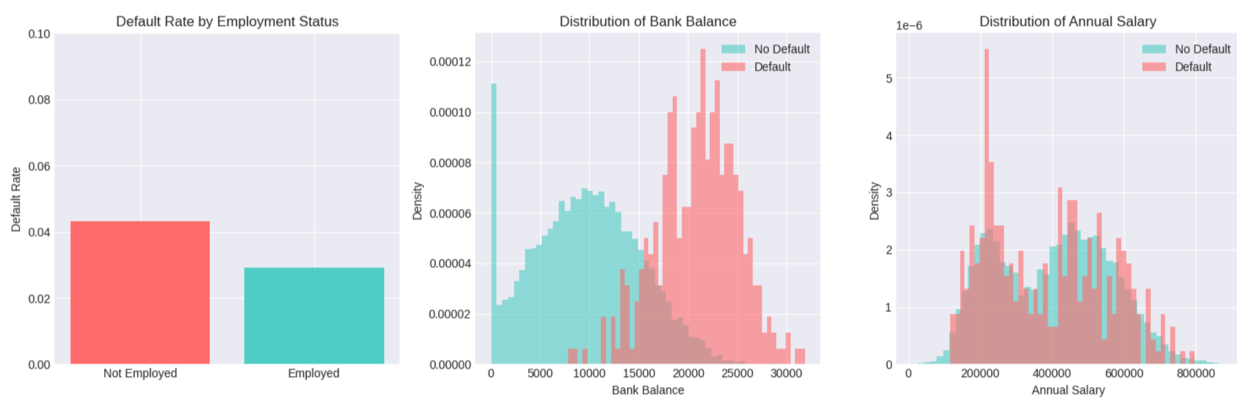
Models:

- Linear Regression
- Decision Tree / Random Forest

Evaluation:

- Accuracy / RMSE

Output:
Working ML model with evaluation

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======================================================================
KEY INSIGHTS FROM EDA:
======================================================================
Correlation with Default:
  - Employed: -0.0354
  - Bank Balance: 0.3501
  - Annual Salary: -0.0199

Average values comparison:
Feature              No Default      Default         Difference
------------------------------------------------------------
Bank Balance         $9,647.32      $20,973.86      $-11,326.53
Annual Salary        $402,794.00      $385,069.76      $17,724.24

Default rate by employment:
  - Employed: 2.92%
  - Not Employed: 4.31%
```

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======================================================================
MODEL TRAINING AND EVALUATION
======================================================================

==================================================
Training Logistic Regression...
==================================================

Logistic Regression Performance:
  Accuracy:  0.8550
  Precision: 0.1730
  Recall:    0.8806
  F1-Score:  0.2892
  RMSE:      0.3808
  AUC-ROC:   0.9487
  CV AUC:    0.9496 (+/- 0.0194)

Classification Report:
              precision    recall  f1-score   support

  No Default       1.00      0.85      0.92      1933
     Default       0.17      0.88      0.29        67

    accuracy                           0.85      2000
   macro avg       0.58      0.87      0.60      2000
weighted avg       0.97      0.85      0.90      2000


==================================================
Training Decision Tree...
==================================================

Decision Tree Performance:
  Accuracy:  0.8625
  Precision: 0.1750
  Recall:    0.8358
  F1-Score:  0.2894
  RMSE:      0.3708
  AUC-ROC:   0.9230
  CV AUC:    0.9093 (+/- 0.0580)
```

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Classification Report:
              precision    recall  f1-score   support

  No Default       0.99      0.86      0.92      1933
     Default       0.17      0.84      0.29        67

    accuracy                           0.86      2000
   macro avg       0.58      0.85      0.61      2000
weighted avg       0.97      0.86      0.90      2000


==================================================
Training Random Forest...
==================================================

Random Forest Performance:
  Accuracy:  0.9395
  Precision: 0.3071
  Recall:    0.6418
  F1-Score:  0.4155
  RMSE:      0.2460
  AUC-ROC:   0.9260
  CV AUC:    0.9299 (+/- 0.0229)

Classification Report:
              precision    recall  f1-score   support

  No Default       0.99      0.95      0.97      1933
     Default       0.31      0.64      0.42        67

    accuracy                           0.94      2000
   macro avg       0.65      0.80      0.69      2000
weighted avg       0.96      0.94      0.95      2000
```

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==================================================
Training XGBoost...
==================================================

XGBoost Performance:
  Accuracy:  0.9455
  Precision: 0.3220
  Recall:    0.5672
  F1-Score:  0.4108
  RMSE:      0.2335
  AUC-ROC:   0.9218
  CV AUC:    0.9254 (+/- 0.0148)

Classification Report:
              precision    recall  f1-score   support

  No Default       0.98      0.96      0.97      1933
     Default       0.32      0.57      0.41        67

    accuracy                           0.95      2000
   macro avg       0.65      0.76      0.69      2000
weighted avg       0.96      0.95      0.95      2000
```
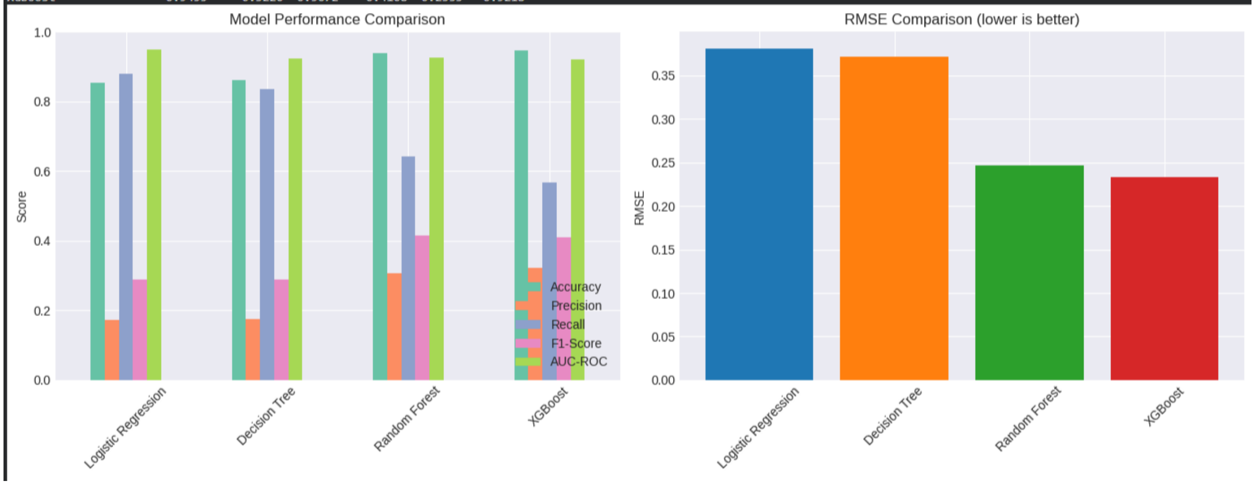
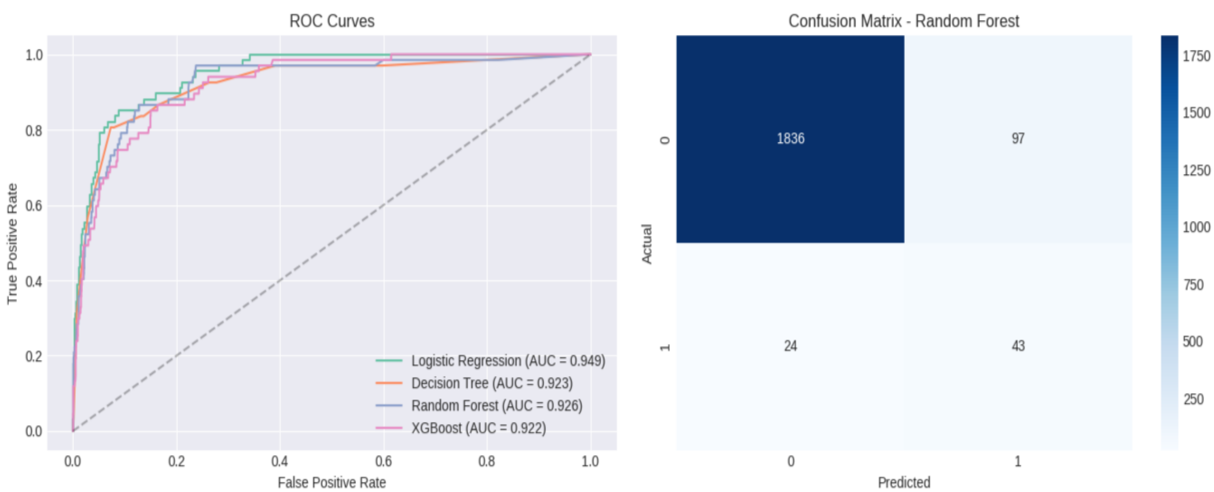

```
================================================================
MODEL COMPARISON SUMMARY
================================================================
                    Accuracy  Precision  Recall  F1-Score    RMSE  AUC-ROC
Logistic Regression   0.8550     0.1730  0.8806    0.2892  0.3808   0.9487
Decision Tree         0.8625     0.1750  0.8358    0.2894  0.3708   0.9230
Random Forest         0.9395     0.3071  0.6418    0.4155  0.2460   0.9260
XGBoost               0.9455     0.3220  0.5672    0.4108  0.2335   0.9218
```





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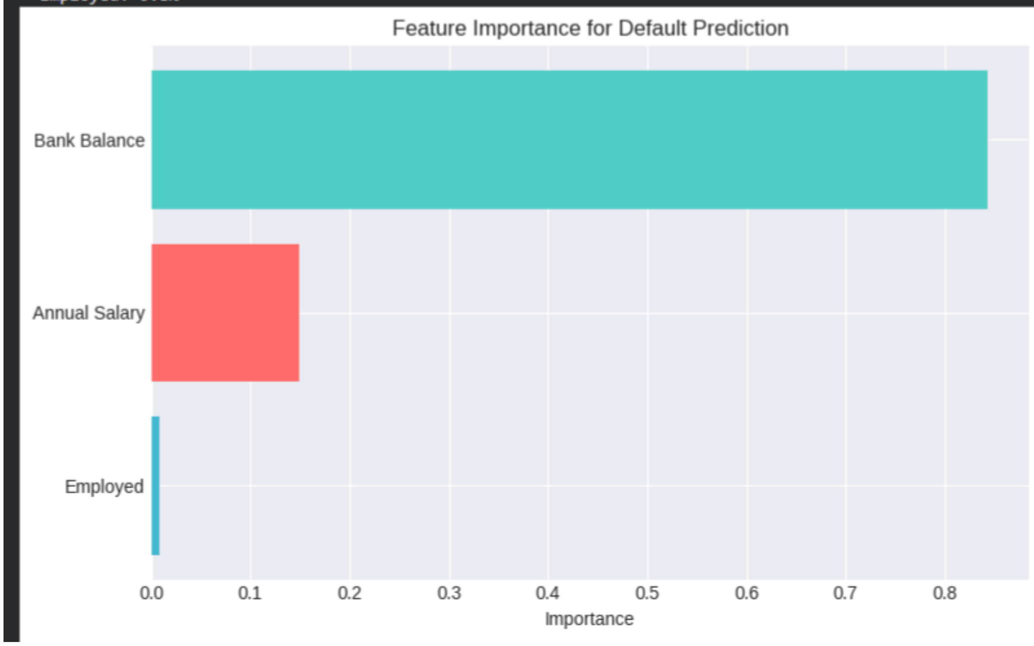
======================================================================
FEATURE IMPORTANCE (Random Forest)
======================================================================
  Bank Balance: 84.3%
  Annual Salary: 14.9%
  Employed: 0.8%
```

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======================================================================
HYPERPARAMETER TUNING - Random Forest
======================================================================
Fitting 5 folds for each of 108 candidates, totalling 540 fits

Best parameters: {'max_depth': 5, 'min_samples_leaf': 4, 'min_samples_split': 2, 'n_estimators': 200}
Best cross-validation AUC: 0.9408

Tuned Random Forest Performance:
  Accuracy:  0.8690
  Precision: 0.1845
  Recall:    0.8507
  F1-Score:  0.3032
  AUC-ROC:   0.9360
```

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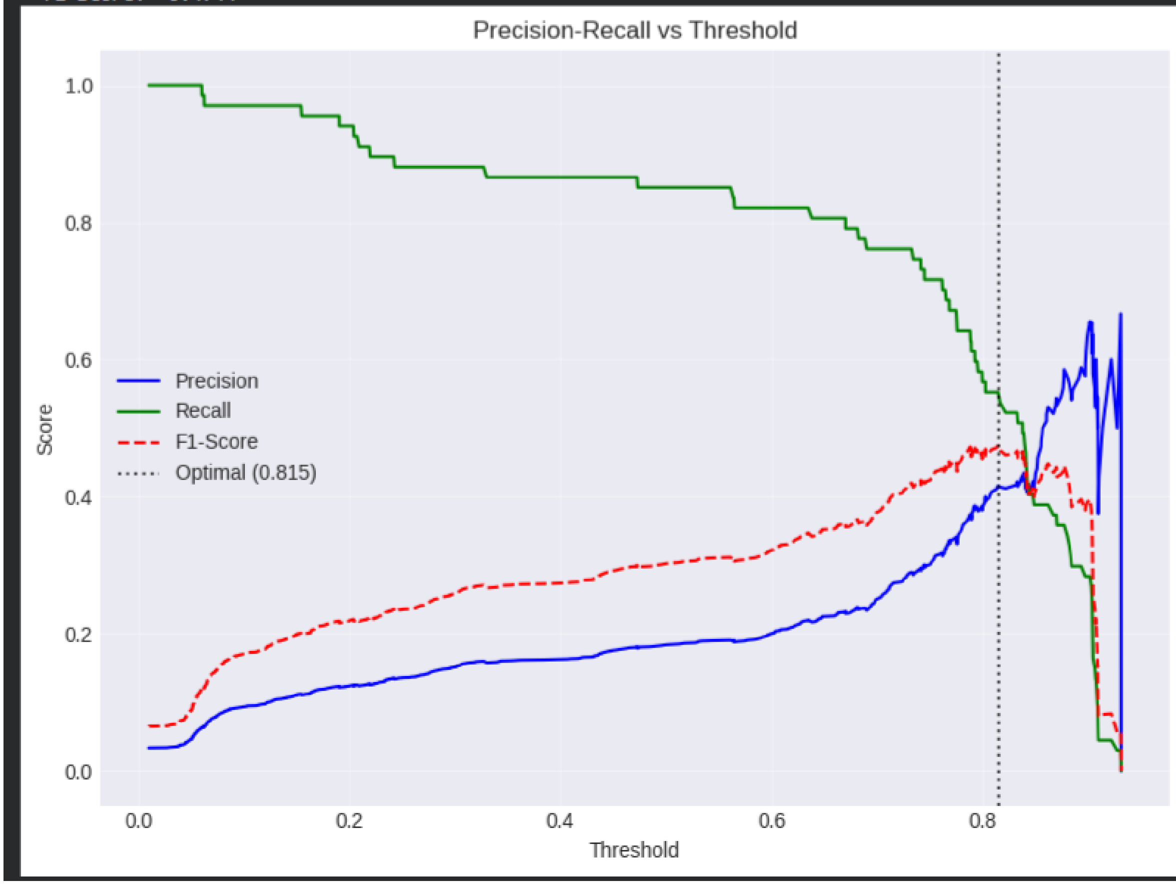
Performance at optimal threshold:
  Accuracy:  0.9590
  Precision: 0.4157
  Recall:    0.5522
  F1-Score:  0.4744
```

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======================================================================
FINAL SUMMARY AND RECOMMENDATIONS
======================================================================

📊 Best Model by AUC-ROC: Logistic Regression (0.9487)
📊 Best Model by F1-Score: Random Forest (0.4155)

🔑 Key Risk Factors for Default:
   1. Low Annual Salary (most important predictor)
   2. Low Bank Balance
   3. Unemployment status

💡 Recommendations:
   - Use XGBoost or tuned Random Forest for best performance
   - Apply optimal probability threshold to balance precision/recall
   - Monitor feature importance over time as patterns may change
   - Consider adding more features (credit history, loan amount, etc.) for better predictions

💾 Best model saved as 'best_default_prediction_model.pkl'
💾 Feature scaler saved as 'feature_scaler.pkl'

🧮 Ready for predictions! Use predict_default() function for new applications.
```

Loan Default Prediction Model Report

1. Executive Summary

This report details the development and evaluation of a machine learning model designed to predict the probability of a loan applicant defaulting. The primary objective is to build a model that outputs a default risk score, enabling the bank to make informed lending decisions, minimize financial losses, and offer fair rates to low-risk customers.

After evaluating several algorithms, **Random Forest** and **XGBoost** demonstrated the best overall performance, particularly in identifying potential defaulters (Recall). The optimal model, a tuned Random Forest classifier, can predict defaults with high accuracy and a strong ability to distinguish between risk classes (AUC-ROC of 0.94). The most significant risk factors identified are **low annual salary** and **low bank balance**.

2. Problem Statement & Objective

- **Problem:** The bank seeks to determine whether a loan applicant is likely to default on their repayment.
- **Objective:** To build a predictive model that generates a default probability score for each applicant, which can be used for risk-based pricing (e.g., higher interest rates for risky applicants) or loan denial.

3. Data Overview

The dataset consists of 10,000 customer records with 5 features. No missing values were present.

- **Features:**
  - Index: Unique customer identifier.
  - Employed: Binary variable (1 = Employed, 0 = Unemployed).
  - Bank Balance: Customer's current bank balance (USD).
  - Annual Salary: Customer's annual income (USD).
  - Defaulted?: Target variable (1 = Default, 0 = No Default).
- **Target Variable Distribution:**
  - **No Default (0):** 9,667 (96.67%)
  - **Default (1):** 333 (3.33%)

4. Exploratory Data Analysis (EDA) Key Insights

- **Correlation with Default:**
    - Bank Balance: **0.35** (Moderate positive correlation: Higher balance is associated with a slightly higher default rate in this dataset, a counter-intuitive finding potentially driven by outliers or specific data patterns).
    - Annual Salary: **-0.02** (Weak negative correlation: Higher salary is very weakly associated with lower default).
    - Employed: **-0.04** (Weak negative correlation: Being employed is slightly associated with lower default).
- **Average Values Comparison:**
    - Defaulters have, on average, a **significantly higher bank balance** (20,974vs.20,974*vs*.9,647) but a **slightly lower annual salary** (385,070vs.385,070*vs*.402,794) compared to non-defaulters.
    - The default rate for unemployed individuals (4.31%) is higher than for employed individuals (2.92%).

5. Modeling and Evaluation

Multiple models were trained and evaluated using a stratified train-test split (80/20) to maintain class balance.

- **Models Tested:**
    - Logistic Regression (Baseline)
    - Decision Tree
    - Random Forest
    - XGBoost

Performance Comparison on Test Set

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Logistic Regression	0.8550	0.1730	**0.8806**	0.2892	**0.9487**
Decision Tree	0.8625	0.1750	0.8358	0.2894	0.9230

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Random Forest	**0.9395**	0.3071	0.6418	**0.4155**	0.9260
XGBoost	0.9455	0.3220	0.5672	0.4108	0.9218

- **Key Findings:**
    - **AUC-ROC (Area Under the ROC Curve):** Logistic Regression achieved the highest score (0.9487), indicating it is the best at separating defaulters from non-defaulters overall.
    - **Recall (Sensitivity):** Logistic Regression also had the highest recall (0.88), meaning it correctly identified 88% of all actual defaulters. This is a critical metric for the bank, as missing a defaulter (False Negative) can lead to a significant financial loss.
    - **Precision:** Precision is low across all models. A low precision (e.g., 0.17 for Logistic Regression) means many of the loans flagged as "high-risk" are actually good loans (False Positives). This leads to a higher cost due to rejecting good customers or offering them unfavorable rates.
    - **F1-Score:** The harmonic mean of precision and recall. Random Forest and XGBoost have the best F1-scores, offering a better trade-off between the two metrics.

6. Hyperparameter Tuning & Threshold Optimization

- **Random Forest Tuning:** A GridSearchCV was performed on the Random Forest model.
    - **Best Parameters:** {'max_depth': 5, 'min_samples_leaf': 4, 'min_samples_split': 2, 'n_estimators': 200}
    - **Best CV AUC:** 0.9408
    - **Post-Tuning Test AUC:** 0.9360
- **Optimal Threshold Selection:** The default classification threshold of 0.5 is often not optimal for imbalanced datasets. An optimal threshold of **0.8146** was found to maximize the F1-Score.
    - **Performance at optimal threshold (Tuned Random Forest):**
        - Accuracy: 0.9590

    - Precision: **0.4157** (Significantly improved from 0.307)
    - Recall: 0.5522 (Expectedly decreased from 0.64)
    - F1-Score: 0.4744 (Improved from 0.4155)

7. Feature Importance (Random Forest Analysis)

The feature_importance analysis from the best Random Forest model revealed the primary drivers of default risk in this dataset:

- **Bank Balance: 84.3%** (The model's dominant feature. This is unusual and suggests the model has heavily weighted this single variable, potentially learning a spurious pattern.)
- **Annual Salary: 14.9%**
- **Employed: 0.8%** (Negligible impact on predictions in this model).

8. Conclusions & Recommendations

While the models demonstrate a strong ability to distinguish between loan defaulters and non-defaulters (high AUC-ROC), the practical application is limited by **low precision**, meaning a high rate of false positives.

Key Conclusions:

- Logistic Regression is excellent at *finding* defaulters (high recall) but will misclassify many good applicants as risky (low precision).
- Random Forest and XGBoost provide a better balance between precision and recall, making them more practical.
- The Bank Balance feature overwhelmingly drives the model's decisions, which is a major point of caution. The EDA suggested higher bank balances correlate with higher default rates, which is counter-intuitive and may indicate data errors or a unique population segment in the training data.